EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 14

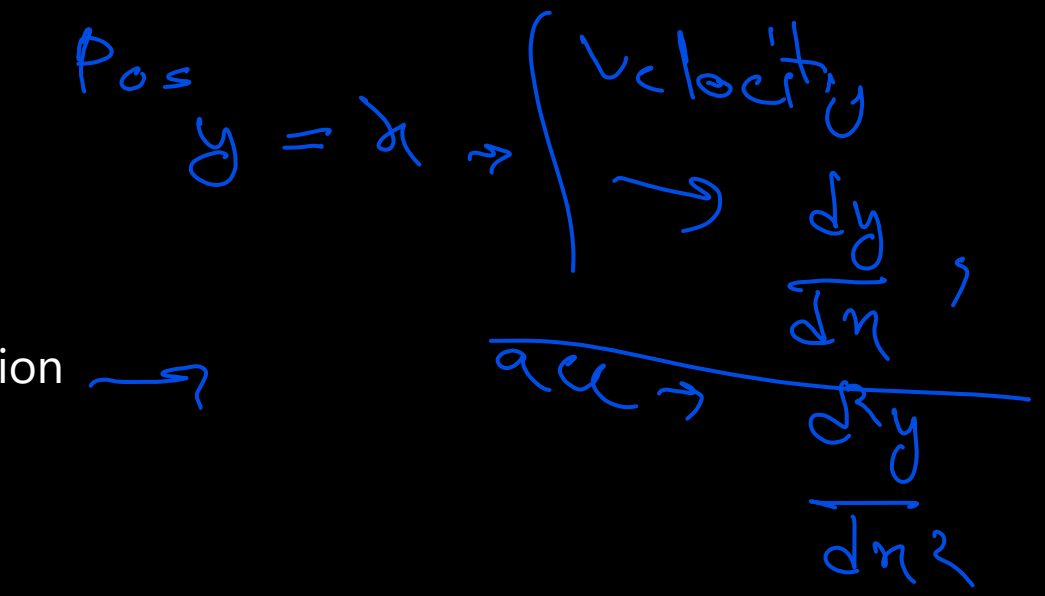
Basic Maths and Calculus

Applications of Differentiation and Maxima-Minima

By - Excellence Programme

18 September 2026

Topics to be covered



01 5-minute differentiation recap

02 Position, velocity and acceleration

03 Exact velocity–time graphs

04 Related rates: circle, square and sphere

05 Stationary points

06 Second derivative test

07 Maxima and minima practice

Rates of change

position x(t) → velocity v = dx/dt →



velocity v(t) → acceleration a = dv/dt = d^2x/dt^2

stationary point: dy/dx = 0

related rates: differentiate both sides with respect to time



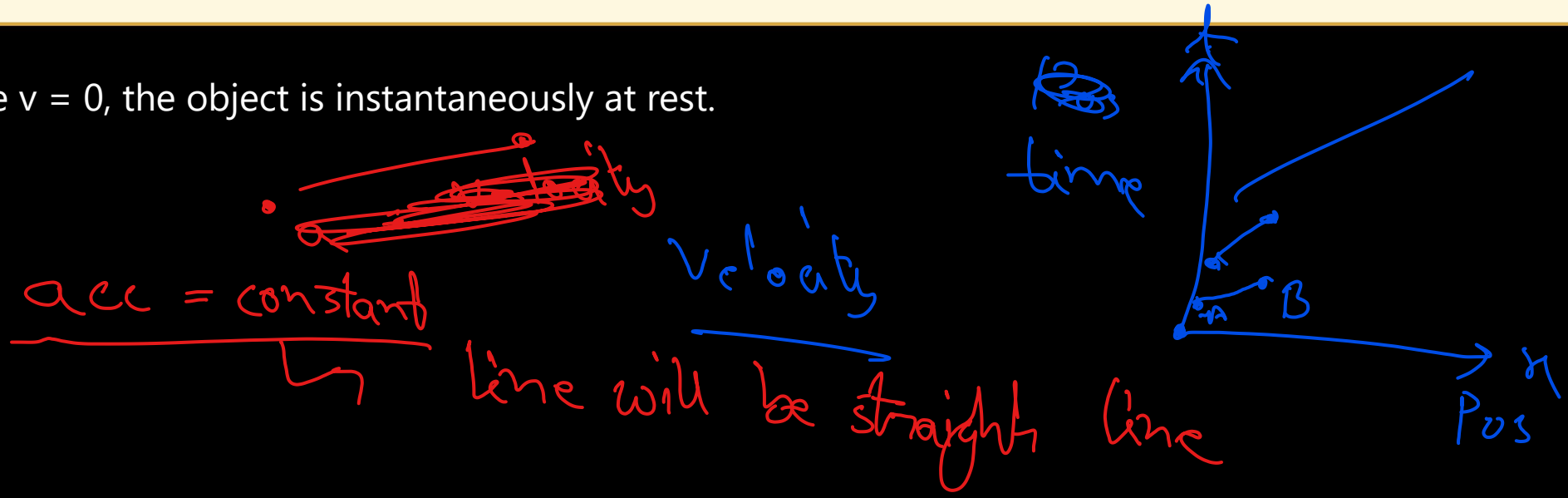
From position to a velocity graph

Differentiate the position function to obtain velocity.

Then plot v against t using its exact equation.

x(t) → v(t) = dx/dt → v–t graph

Where v = 0, the object is instantaneously at rest.



Motion: increasing velocity line

If $x = t^2-4t+8$, find velocity and draw the exact v–t graph.

position → velocity = ?

draw v-t graph

$v = \frac{dx}{dt} = 2t - 4$

Plot v-t graph

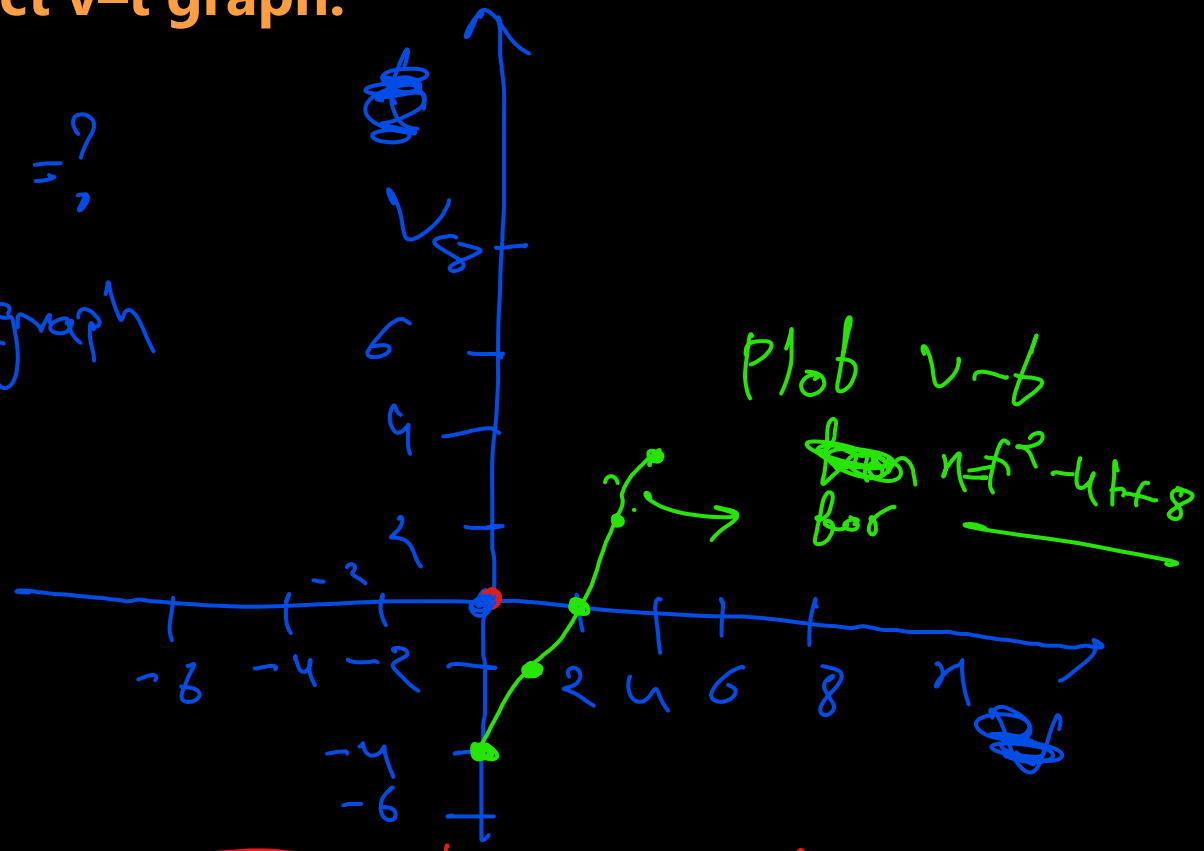
If $t = 0 \rightarrow v = -4$

$t = 1, -2$

$t = 2 \Rightarrow 0$

$t = 3 \rightarrow 2$

$t = 4, 4$



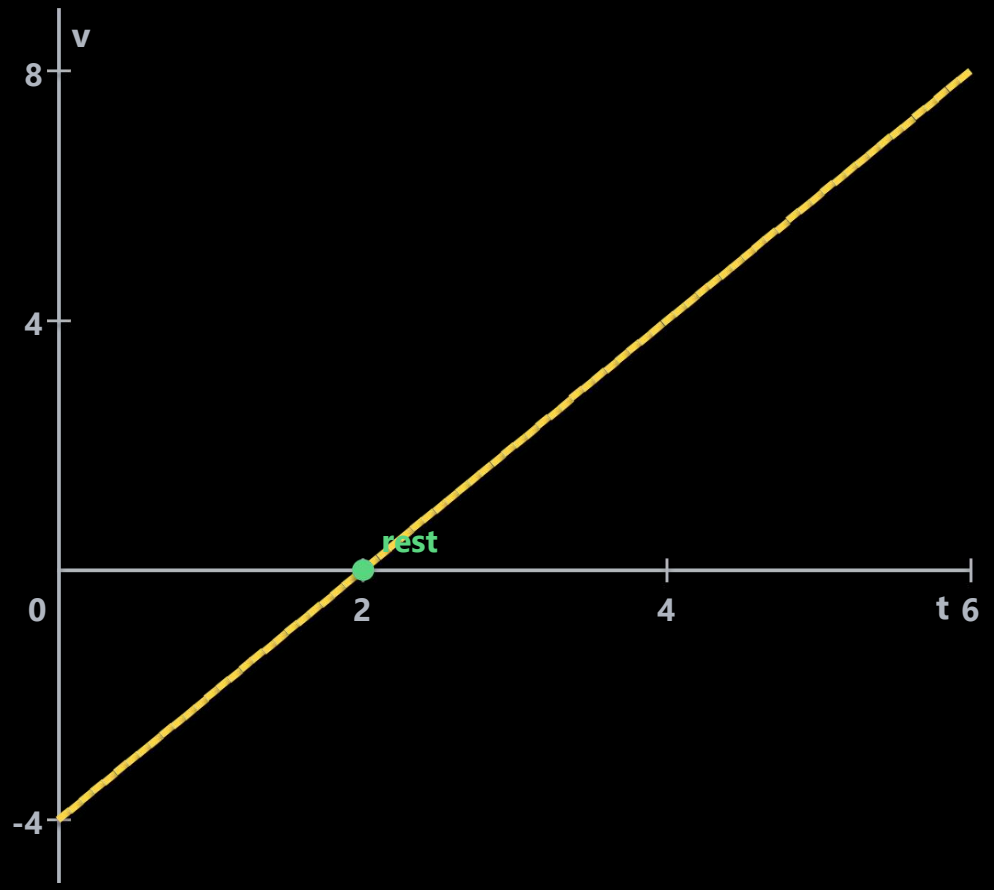
Plot v-t for $x = t^2 - 4t + 8$

0	1	2	3	4
-4	-3	0	2	4

Solution: Motion: increasing velocity line

Step 1: v = dx/dt = 2t-4

Step 2: v(0) = -4 and v = 0 at t = 2



Rough work / extra work

Motion: decreasing velocity line

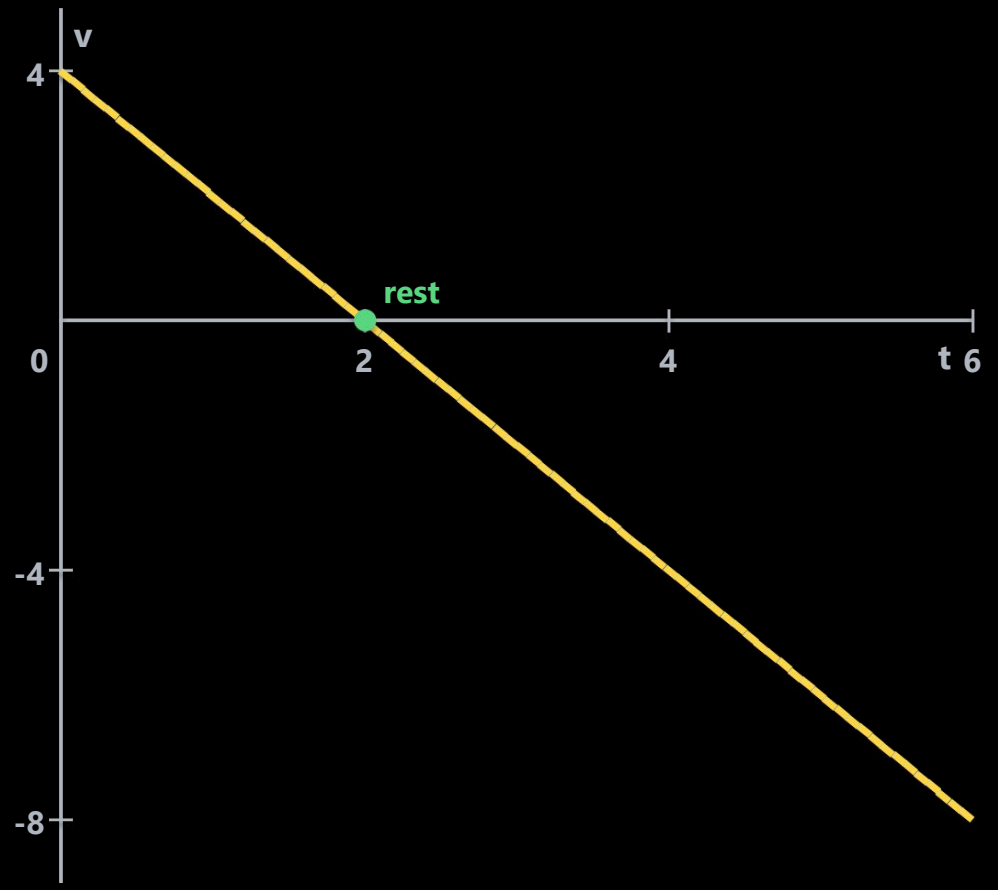
H.W

If $x = -t^2+4t+8$, find velocity and draw the exact v–t graph.

Solution: Motion: decreasing velocity line

Step 1: v = dx/dt = -2t+4

Step 2: v(0) = 4 and v = 0 at t = 2



Acceleration from velocity

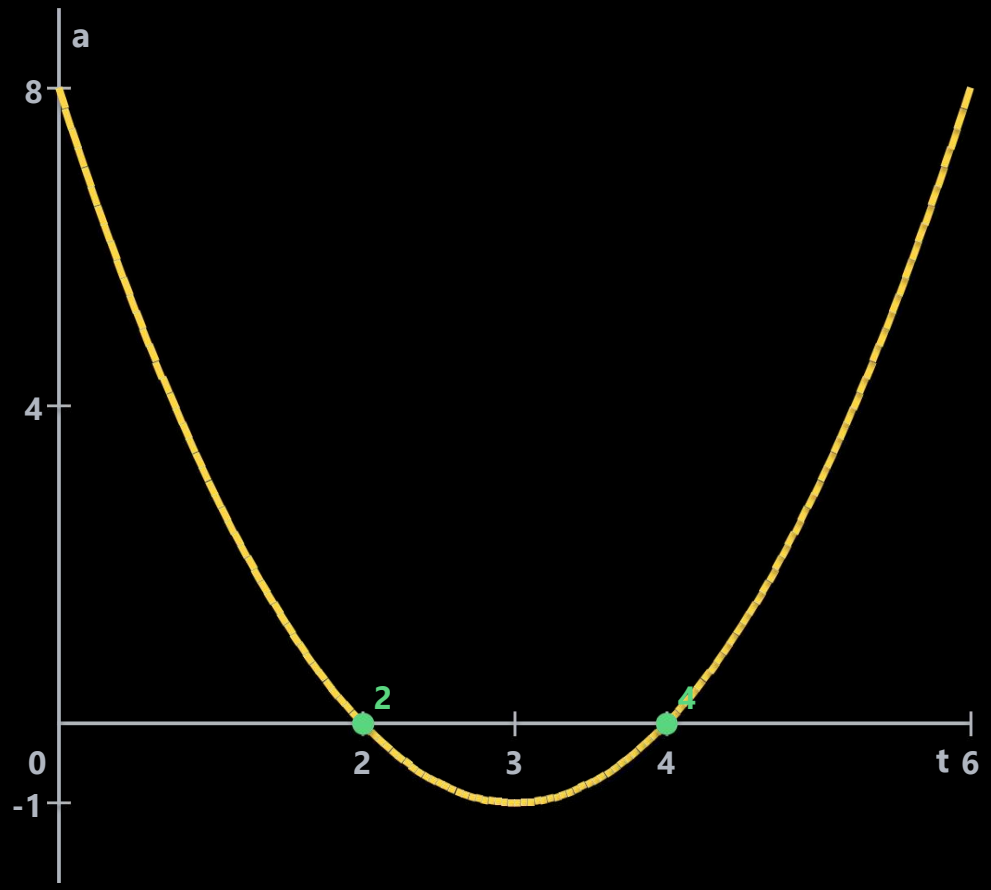
H.W

If $v = t^3/3-3t^2+8t$, find acceleration and draw the exact a–t graph.

Solution: Acceleration from velocity

Step 1: $a = dv/dt = t^2-6t+8$

Step 2: $a = (t-2)(t-4)$



Rough work / extra work

Changing x with time

If $y = x^2$ and x changes with time, express dy/dt in terms of x and dx/dt.

$y = x^2$ → If x will change It will affect value of y

x → change with time



$$\frac{dy}{dx} = 2x$$

$$\frac{\frac{dy}{dt}}{\frac{dx}{dt}} = 2x \rightarrow$$

$$\boxed{\frac{dy}{dt} = 2x \times \frac{dx}{dt}}$$

Solution: Changing x with time

Step 1: $dy/dx = 2x$

Step 2: $dy/dt = (dy/dx)(dx/dt)$

$dy/dt = 2x(dx/dt)$.

Changing angle with time

If y = sin θ and θ changes with time, express dy/dt. → in term of θ & $\frac{d\theta}{dt}$

$y = \sin\theta$; $\frac{dy}{d\theta} = \cos\theta$,

$\frac{\frac{dy}{dt}}{\frac{d\theta}{dt}} = \cos\theta \rightarrow$

$\frac{dy}{dt} = \frac{d\theta}{dt} \times \cos\theta$

Solution: Changing angle with time

Step 1: $dy/d\theta = \cos\theta$

Step 2: $dy/dt = (dy/d\theta)(d\theta/dt)$

$dy/dt = \cos\theta(d\theta/dt)$.

Nested time dependence

H.W

If $y = \sin(t^2+2t)$, find dy/dt.

Solution: Nested time dependence

Step 1: Outer derivative: $\cos(t^2+2t)$

Step 2: Inner derivative: $2t+2$

$dy/dt = (2t+2)\cos(t^2+2t)$.

Rough work / extra work

Related rates

When dimensions change with time, geometrical quantities also change.

Write the geometry formula first, then differentiate with respect to time.

formula → differentiate with respect to t → substitute values with units

Substitute numerical values only after differentiating.

position change with ~~time~~ ⇒ m/s (velocity)

↳ In meter

↳ Quantity is changed from distance to velocity

Circle, square and sphere formulae

In a circle when radius change with time → Area will be change with time → r

circle: $A = \pi r^2 \rightarrow dA/dt = 2\pi r(dr/dt)$

square: $P = 4l \rightarrow dP/dt = 4(dl/dt)$

sphere: $V = (4/3)\pi r^3 \rightarrow dV/dt = 4\pi r^2(dr/dt)$

sphere: $S = 4\pi r^2 \rightarrow dS/dt = 8\pi r(dr/dt)$

area of circle $(A) = \pi r^2 = A$

$\frac{dr}{dt}$ → Exist | We have find $\frac{dA}{dt} = ?$

Step-1) $\frac{dA}{dr} \Rightarrow \frac{d(\pi r^2)}{dr}$

$\Rightarrow \pi \frac{dr^2}{dr} \Rightarrow 2\pi r$

$\frac{dA}{dr} = 2\pi r$

$\frac{dA}{dt} \Big/ \frac{dr}{dt} \Rightarrow = 2\pi r \Rightarrow \frac{dA}{dt} = 2\pi r \times \frac{dr}{dt}$

Sphere volume rate

H.W

For $V = (4/3)\pi r^3$, express dV/dt in terms of r and dr/dt.

Solution: Sphere volume rate

Step 1: Differentiate with respect to t

$dV/dt = 4\pi r^2(dr/dt)$.

Rough work / extra work

Sphere surface-area rate

Solve It

For $S = 4\pi r^2$, express dS/dt in terms of r and dr/dt.

$$\frac{dS}{dt} \rightarrow \frac{dS}{dr} \Rightarrow 4\pi \times 2r \Rightarrow 8\pi r$$

$$\frac{dS}{dr} = 8\pi r$$

$$\frac{\frac{dS}{dt}}{\frac{dr}{dt}} = 8\pi r \Rightarrow$$

$$\boxed{\frac{dS}{dt} = 8\pi r \times \frac{dr}{dt}}$$

Solution: Sphere surface-area rate

Step 1: $d(r^2)/dt = 2r(dr/dt)$

$dS/dt = 8\pi r(dr/dt)$.

Expanding circle

A circle's radius increases at $1/\pi$ m s^{-1}. Find dA/dt when r = 4 m.

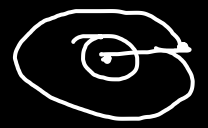
$\frac{dA}{dt}$ at $r = 4$ m

$\frac{dr}{dt} = \frac{1}{\pi}$ | $A = \pi r^2$

$\frac{dA}{dr} = \frac{d\,\pi r^2}{dr} = 2\pi r$

$\frac{\frac{dA}{dt}}{\frac{dr}{dt}} = 2\pi r \Rightarrow 2\pi r \times \frac{dr}{dt}$

$\Rightarrow 2\pi r \times \frac{1}{\pi} \Rightarrow 2r$ & At $r = 4 \Rightarrow 2 \times 4 \Rightarrow 8$



On each second Radius will be increased by $1/\pi$ m s^{-1}

Solution: Expanding circle

Step 1: $dA/dt = 2\pi r(dr/dt)$

Step 2: $= 2\pi(4)(1/\pi)$

$dA/dt = 8\ m^2\ s^{-1}$.

Growing square perimeter

The side of a square increases at 0.2 cm s^{-1}. Find the rate of increase of its perimeter.

Solution: Growing square perimeter

Step 1: $P = 4l$

Step 2: $dP/dt = 4(dl/dt) = 4(0.2)$

$dP/dt = 0.8\ \text{cm s}^{-1}$.

Rough work / extra work

Area change with radius

Solution

For a circle $A = \pi r^2$, find dA/dr.

$$\frac{dA}{dr} = A = \pi r^2$$

$$\Rightarrow \frac{d}{dr}(\pi r^2) \Rightarrow \pi \frac{dr^2}{dr} = \pi \times 2r \Rightarrow 2\pi r$$

Solution: Area change with radius

$dA/dr = 2\pi r$.

Inflating balloon

A spherical balloon's volume increases at 10 cm³ s⁻¹. Find dr/dt when r = 5 cm.

$\frac{dV}{dt} = 10$ | $\frac{dr}{dt} = ?$, $r = 5$

$V = \frac{4}{3}\pi r^3$

$\frac{dV}{dr} = \frac{4}{3}\pi \frac{dr^3}{dr} \Rightarrow \frac{4}{3}\pi \times 3r^2$

$\frac{dV}{dr} = 4\pi r^2$

$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$

$10 = 4\pi r^2 \times \frac{dr}{dt}$

$\frac{dr}{dt} = \frac{10}{4\pi r^2}$

$= \frac{5}{2\pi r \times 5R}$

$\frac{dr}{dt} = \frac{1}{10\pi}$

Ans

Solution: Inflating balloon

Step 1: $dV/dt = 4\pi r^2(dr/dt)$

Step 2: $10 = 4\pi(25)(dr/dt)$

$dr/dt = 1/(10\pi)$ cm s^{-1}.

Rough work / extra work

Growing square area

A square's side increases at 2 cm s^{-1}. Find dA/dt when the side is 10 cm.

Solution: Growing square area

Step 1: $A = l^2 \rightarrow dA/dt = 2l(dl/dt)$

Step 2: $= 2(10)(2)$

$dA/dt = 40\ cm^2\ s^{-1}$.

When is the particle at rest?

Solve

If $v(t) = t^2 - 3t$, when is the particle at rest?

time when $\frac{dx}{dt}$ or velocity becomes zero

If $t = 0$, $0 - 3 \times 0 = 0$

$t = 3$, $3 - 3 \times 3 = 0$

$v = 0$

$t^2 - 3t = 0$

$t(t-3) = 0$

$t = 0, t = 3$

Solution: When is the particle at rest?

Step 1: At rest: $v = 0$

Step 2: $t(t-3) = 0$

$t = 0$ s or 3 s.

Acceleration from displacement

Sure If

If $s(t) = 3t^2-4t+2$, find acceleration at t = 1 s.

↳ s displacement ↳

$v = \frac{ds}{dt} = 6t - 4$

$a = \frac{dv}{dt} = 6$

$a = 6$ →

$t = 1, 2, 3, \ldots$

a will be always 6 (constant)

Solution: Acceleration from displacement

Step 1: $v = ds/dt = 6t - 4$

Step 2: $a = dv/dt = 6$

Acceleration = 6 m s^{-2}.

Rough work / extra work

Velocity from displacement

H.W

If $x(t) = 4t^3+3t^2-6t+1$, find velocity at $t = 2$ s.

Solution: Velocity from displacement

Step 1: $v = dx/dt = 12t^2 + 6t - 6$

Step 2: $v(2) = 48 + 12 - 6$

Velocity = 54 m s^{-1}.

Exponential position

Position given & find velocity

If $x(t) = e^{2t}+5$, find velocity.

$$v = \frac{dx}{dt} = 2 \times e^{2t}$$

Good

Solution: Exponential position

$v = dx/dt = 2e^{2t}$.

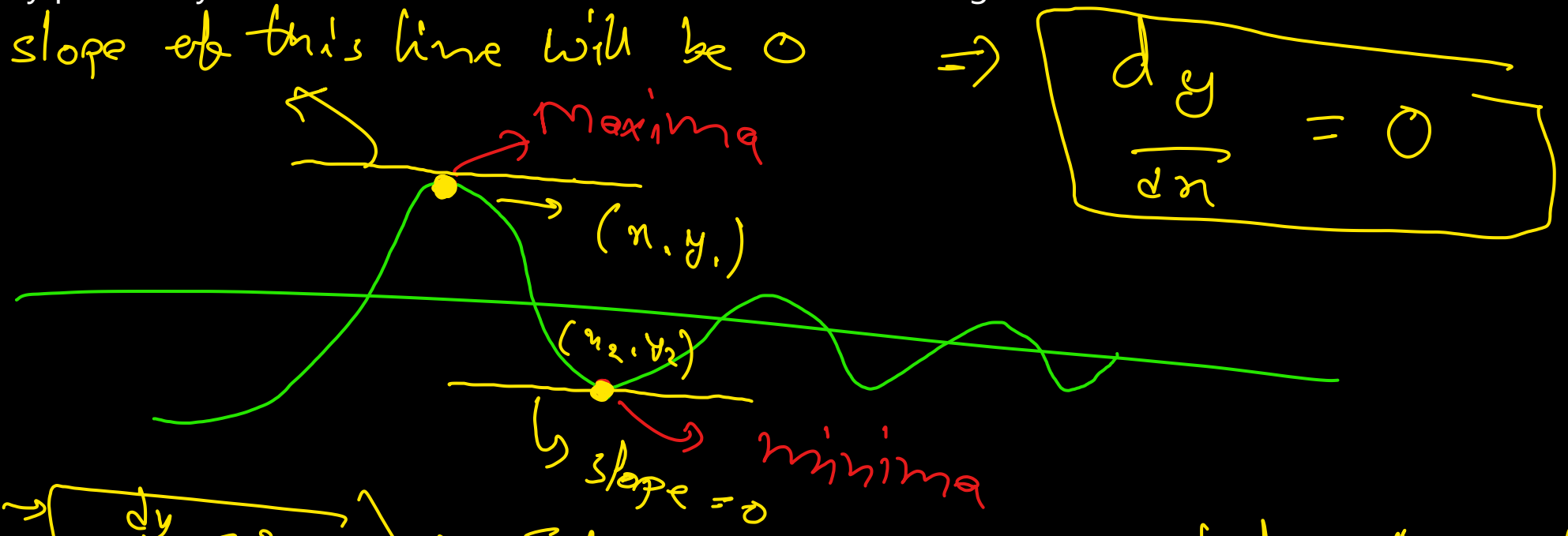
Rough work / extra work

Maxima and minima

At a smooth local maximum or minimum, the tangent is horizontal.

Step 1: solve dy/dx = 0 to find stationary points

A stationary point may be a maximum, minimum or neither. Test it using the second derivative.



Second derivative test

$d^2y/dx^2 > 0$ at a stationary point → local minimum

$d^2y/dx^2 < 0$ at a stationary point → local maximum

If $d^2y/dx^2 = 0$, this test is inconclusive; another test is needed.

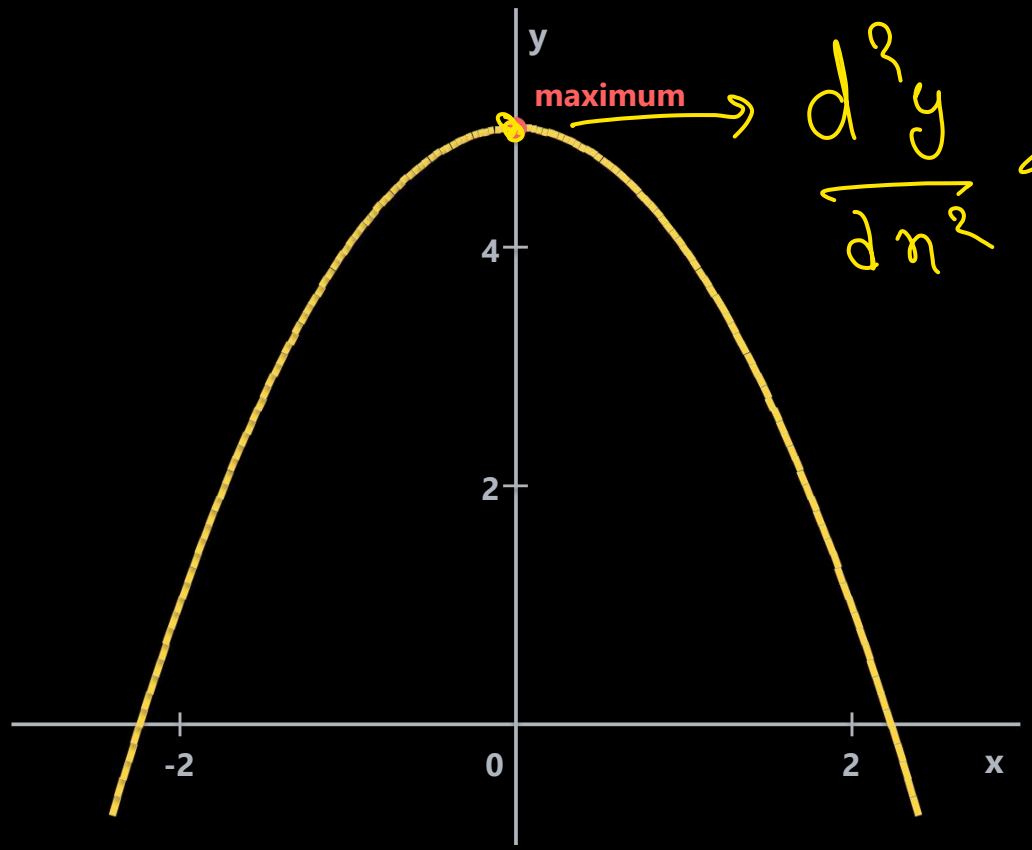
$y'|_{x_1,y_1}$ → check If > 0 → It will local minima

↳ If < 0 ⇒ It will be local maxima

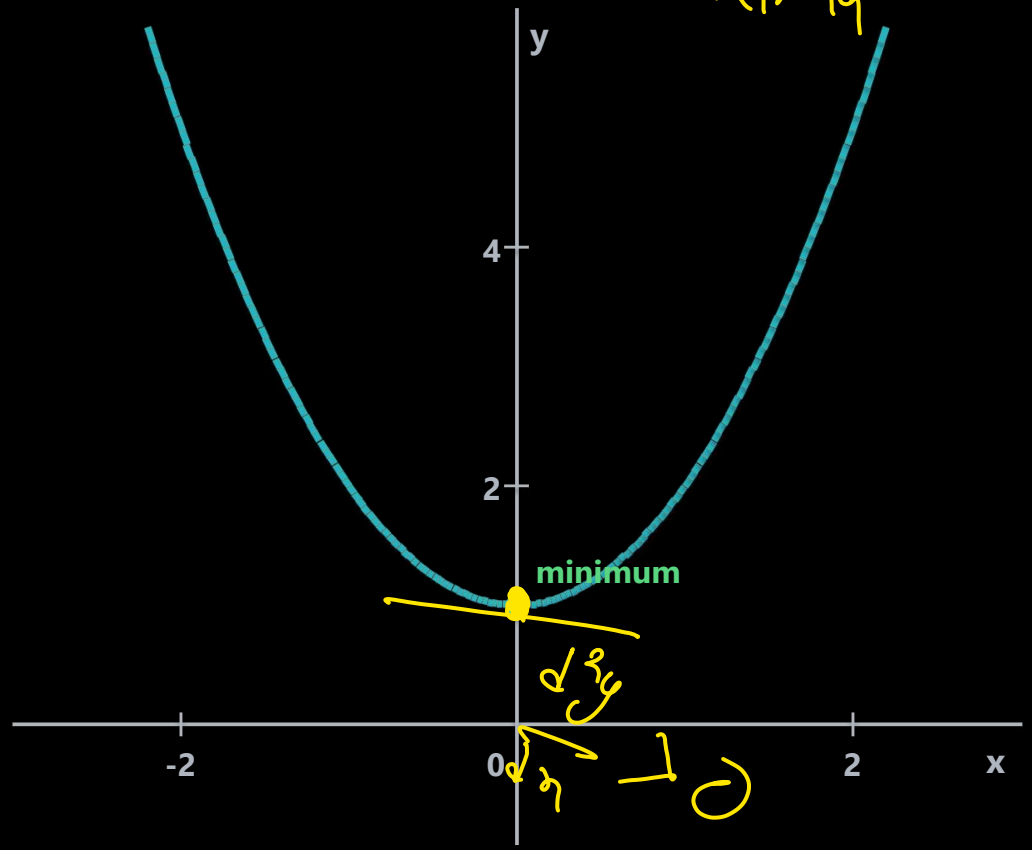
Maximum and minimum shapes

If $\frac{d^2y}{dx^2} > 0$ → Then minima

→ < 0 → Then maxima



$\frac{d^2y}{dx^2} < 0$



$\frac{d^2y}{dx^2} > 0$

Minimum of a quadratic

Solve

Find the maxima and minima of $y = x^2+5$.

minima = 5

Maxima → 0

$\frac{dy}{dx} = 0 \Rightarrow 2x = 0 \Rightarrow x = 0$

$\frac{d^2y}{dx} = 2 \Rightarrow > 0 \Rightarrow$ minima

at $x = 0$ → minima Exist

Put $x = 0$ in $y = x^2 + 5 \Rightarrow 5$ → minima

Maxima not Exist

9

Solution: Minimum of a quadratic

Step 1: $dy/dx = 2x = 0 \rightarrow x = 0$

Step 2: $d^2y/dx^2 = 2 > 0 \rightarrow$ minimum

Minimum value y = 5 at x = 0; no maximum.

$\frac{d^2y}{dx^2} < 0$ (This way not found)

in our case

so no maxima Exist

Rough work / extra work

Maximum of a quadratic

H, L

Find the maxima and minima of $y = 4x-x^2$.

Solution: Maximum of a quadratic

Step 1: $dy/dx = 4-2x = 0 \rightarrow x = 2$

Step 2: $d^2y/dx^2 = -2 < 0 \rightarrow$ maximum

Maximum value $y = 4$ at $x = 2$; no minimum.

Two stationary points

max at 0 & Ih y
min at 2 & Ih is 0

Solve It

For $y = x^3-3x^2+4$, find its local maximum and minimum values.

max at x 0	+ ~~3~~	↗
& min at 2	− 6	↘
		0

$y = x^3 - 3x^2 + 4$

$\frac{dy}{dx} = 3x^2 - 6x = 0$

$\Rightarrow 3x(x-2) = 0$

$x = 0, 2$

$\frac{d^2y}{dx} = 6x - 6$

at $x = 0 \Rightarrow -6$

< 0 which

So maxima at 0

at $x = 2 \Rightarrow 6$

$> 0 \Rightarrow$ minima at $x = 2$

max at $x = 0$

y

min at $x = 2$

$8 - 12 + 4$

$= 0$ min

Solution: Two stationary points

Step 1: $dy/dx = 3x(x-2) = 0 \rightarrow x = 0, 2$

Step 2: $d^2y/dx^2 = 6x-6$

Step 3: $x = 0$ gives maximum; $x = 2$ gives minimum

Local maximum = 4; local minimum = 0.

Maximum and minimum velocity

H. W

If $v = t^3-6t^2+12$, find its local maximum and minimum values.

Solution: Maximum and minimum velocity

Step 1: $dv/dt = 3t(t-4) = 0 \rightarrow t = 0, 4$

Step 2: $d^2v/dt^2 = 6t-12$

Local maximum $v = 12$ at $t = 0$; local minimum $v = -20$ at $t = 4$.

Rough work / extra work

Minimum by completing the test

H.W

Find the maxima and minima of $y = x^2-4x+5$.

Solution: Minimum by completing the test

Step 1: $dy/dx = 2x-4 = 0 \rightarrow x = 2$

Step 2: $d^2y/dx^2 = 2 > 0$

Minimum value $y = 1$ at $x = 2$; no maximum.

Cubic maxima and minima

For $y = x^3-12x^2+36x$, find the local maximum and minimum values.

So for max'i

$$\frac{dy}{dx} = 3x^2 - 24x + 36 = 0$$

$$x(x^2 - 8x + 12) = 0$$

$$x(x^2 - 6x - 2x + 12) = 0$$

$$x(x(x-6) - 2(x-6)) = 0$$

$$x \times (x-2) \times (x-6)$$

$x = 0, 2, 6$

$$\frac{d^2y}{dx^2} = 6x - 24$$

at $x=0 \Rightarrow -24 \Rightarrow$ maxima

at $x=2 \Rightarrow -12 \Rightarrow$ max

at $x=6 \Rightarrow 12 \Rightarrow$ min

local minima

36

6

21 6

local max

$x = 0$

$0 - 0 + 0 = 0$

$x = 2$

$8 - 48 + 36$

-4

$x = 6$

$216 - 432 + 216$

$= 0$

Solution: Cubic maxima and minima

Step 1: $dy/dx = 3(x-2)(x-6) = 0 \rightarrow x = 2, 6$

Step 2: $d^2y/dx^2 = 6x-24$

Local maximum = 32 at x = 2; local minimum = 0 at x = 6.

Rough work / extra work

For maxima & minima points we find point where

$\frac{dy}{dx} = 0$ | Then find $\frac{d^2y}{dx^2}$ ⟶ > 0 ⇒ minima

< 0 ⇒ maxima

Then put value of x of minima & maxima in y

Applications of differentiation

$v = dx/dt$, $a = dv/dt = d^2x/dt^2$

related rates: differentiate the geometry formula with respect to t

stationary point: $dy/dx = 0$

$d^2y/dx^2 > 0$ minimum; $d^2y/dx^2 < 0$ maximum

Practice

1. Redraw all three motion graphs using their exact equations.

2. Memorise the related-rate formulae for a circle, square and sphere.

3. Redo all maxima–minima questions without seeing the answers.

4. Include units in every rate answer.

Never call dy/dx = 0 alone proof of a maximum or minimum.